

Tracing Sparse Causal Features in an Open Language Model

A Small-Scale Reproducibility Study of Attribution Graphs

MPhil in Data Intensive Science

Project 28: Mechanistic Interpretability of Open Source LLMs

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Abstract

Anthropic’s *On the Biology of a Large Language Model* combines sparse features, attribution graphs and causal interventions to formulate and test hypotheses about language-model computation. This study investigates whether that workflow transfers to the open-weight, four-billion-parameter **Qwen3-4B-Instruct-2507** under a deliberately small-scale design. Two behaviours were examined: two-digit addition, with emphasis on decimal carry propagation, and factual recall, instantiated by capital-city and SI-unit prompts. An independent pipeline generated synthetic prompt corpora, captured MLP outputs at seven transformer layers, trained behaviour-specific sparse autoencoders (SAEs), constructed contrastive attribution graphs, and evaluated graph features by inhibition and swap-in interventions. Complete-latent and raw-MLP patches served as positive controls.

The investigation proceeded through successive falsification tests. Initial ReLU SAEs reconstructed held-out activations well (mean validation fraction of variance explained 0.907–0.959), but activated 2,167–3,545 of 8,192 latents on average. Thus poor sparse control could not be attributed simply to reconstruction error; the learned codes were insufficiently sparse. Broad patches confirmed that relevant causal information was present. Initial graph-derived and predeclared TopK arithmetic panels failed matched no-carry specificity tests. A subsequent output-digit-balanced analysis nevertheless localised carry information to late MLP outputs, and inhibition of a frozen 20-feature panel produced a smaller carry-selective effect that replicated on 32 intervention-untouched cases (paired effect -0.086 logits, bootstrap 95% interval $[-0.137, -0.039]$). No answer token flipped. Capital-city swaps remained weak even after replacing low-confidence prompts with high-confidence examples.

The strongest effect remained the predeclared SI-unit TopK follow-up. The selected $k = 128$ SAE retained mean validation FVE 0.936 while reducing mean activity to 99.8 latents. Features were ranked on eight discovery systems and a frozen ten-feature panel was tested on 16 disjoint confirmation systems. Swapping force-source values into energy targets increased the newtons-minus-joules logit gap by 1.234 logits (bootstrap 95% interval $[1.148, 1.332]$), whereas a matched mass-source control changed it by -0.090 ($[-0.129, -0.051]$). The paired specificity effect was 1.324 logits ($[1.223, 1.430]$) and had the predicted sign in all 16 cases. The top prediction nevertheless remained joules. The results support compact, distributed carry- and force-associated causal panels, not complete circuits, monosemantic variables or answer flips. Capital-city controls delimit the transferability of the method.

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1 Introduction

Language models can produce correct outputs without revealing the internal computations that generated them. Mechanistic interpretability seeks causal accounts of those computations in terms of internal representations and their interactions. A central obstacle is superposition: a model may encode more features than it has individual neurons, so a neuron need not correspond to one human-interpretable concept [5]. Sparse autoencoders (SAEs) attempt to recover a more useful basis by reconstructing model activations from a small number of active learned directions [3, 4].

Lindsey et al. applied this programme at scale in *On the Biology of a Large Language Model* [1]. Their attribution graphs trace locally important paths from prompt tokens through sparse features to output logits, yielding hypotheses about mechanisms such as addition and factual recall. The companion methods study emphasises that an attribution graph is a local replacement model and a hypothesis generator; causal interventions in the underlying model remain necessary [2]. This qualification is fundamental to reproducibility. A visually plausible graph is not itself a reproduced mechanism, and a negative intervention is scientifically informative when the implementation and controls distinguish competing explanations.

The present study independently implements this sequence for **Qwen3-4B-Instruct-2507**, an open-weight causal language model with 4.0 billion parameters, 36 transformer layers and hidden width 2,560 [7, 8]. The target is methodological rather than architectural reproduction: small synthetic prompt sets, lightweight per-layer SAEs, a pruned attribution-style graph, and validation by inhibition or swap-in interventions. The two principal behaviours are addition and factual recall. Factual recall is examined both through capital-city prompts, which parallel the target paper’s narrative, and SI units, which constitute a science-focused extension and permit controlled minimal pairs such as force, energy and mass.

Three research questions organise the analysis:

1. Can a lightweight per-layer SAE pipeline identify compact feature sets whose interventions alter a pre-specified output contrast?
2. Do such effects generalise beyond the graph prompt and distinguish the hypothesised concept from matched alternatives?
3. When a sparse intervention fails, do reconstruction diagnostics and broader activation controls identify where the reproducibility chain breaks?

The principal contribution is an evidence chain rather than a binary claim of success or failure. Initial sparse interventions were weak; broad patches then established that causal information was present; reconstruction diagnostics identified excessive latent density; and a fixed TopK comparison tested that diagnosis. The revised representation yielded a compact and specific unit-domain effect. Initial arithmetic controls rejected graph-derived and ten-feature carry interpretations, but a stricter output-digit-balanced analysis subsequently produced a smaller 20-feature effect that survived a frozen case-level replication. This progression permits a narrower and stronger conclusion than either presenting only positive results or treating all weak interventions as an undifferentiated failure.

2 Background and relation to the target study

2.1 Sparse activation decompositions

Let $x \in \mathbb{R}^d$ denote an MLP output and let $u = x/s$ be its normalised form. The initial SAE encodes and reconstructs u as

$$z = \text{ReLU}(W_e(u - b_d) + b_e), \quad (1)$$

$$\hat{u} = W_d z + b_d. \quad (2)$$

Here $m = \dim(z) > d$, so sparsity rather than dimensional compression is intended to make decoder columns candidate feature directions. Reconstruction fidelity is measured by the fraction of variance explained,

$$\text{FVE} = 1 - \frac{\sum_n \|u_n - \hat{u}_n\|_2^2}{\sum_n \|u_n - \bar{u}_{\text{train}}\|_2^2}. \quad (3)$$

High FVE establishes that the dictionary preserves typical activations; it does not establish monosemanticity, sparse causal control, or faithful behaviour under off-manifold edits.

The follow-up models use a hard TopK activation,

$$z = \text{TopK}_k(\text{ReLU}(W_e(u - b_d) + b_e)), \quad (4)$$

which retains only the k largest positive coordinates and sets the remainder to zero. This directly controls per-example support size and is an established alternative to tuning an ℓ_1 coefficient [6]. In the present implementation decoder columns are projected to unit norm after each optimiser step. This removes a scale ambiguity relevant to the original ReLU models and makes the size of a latent intervention more interpretable.

2.2 Anthropic’s attribution graphs

The target study used a 30-million-feature cross-layer transcoder (CLT) for Claude 3.5 Haiku rather than independent per-layer output SAEs [1]. In simplified notation, a feature at layer r reads the residual stream and may contribute to multiple subsequent MLP outputs:

$$a^r = \text{JumpReLU}(W_{\text{enc}}^r x^r), \quad (5)$$

$$\hat{y}^\ell = \sum_{r \leq \ell} W_{\text{dec}}^{r \rightarrow \ell} a^r. \quad (6)$$

Features across layers are therefore trained jointly. For a particular prompt, Anthropic’s local replacement model restores observed reconstruction errors and freezes selected attention and normalisation quantities so that the unperturbed forward pass is reproduced exactly. The resulting graph remains a local approximation; intervention and perturbation-faithfulness tests determine whether it responds like the underlying model away from the clean point [2].

Table 1 states the correspondence precisely. The present work reproduces the scientific workflow at a smaller scale, but it does not treat independent output SAEs as architecturally equivalent to a CLT. This distinction is used when interpreting both the successful units panel and the negative arithmetic result.

Table 1: Methodological correspondence to the target study.

Component	Anthropic target study	Present study
Model	Claude 3.5 Haiku and a smaller research model	Open-weight <code>Qwen3-4B-Instruct-2507</code>
Dictionary	Joint CLT; 30 million features for Haiku	Seven independent MLP-output SAEs; 8,192 latents each
Training support	All model layers, jointly trained on large corpora	1,000 behaviour-specific final-token activations at layers 4, 8, 12, 16, 20, 24 and 28
Local graph	Explicit error nodes; selected operations frozen	Clean-value straight-through splice; gradients through the original model
Feature support	Features across layers and token positions	SAE features at the final prompt position
Validation	Constrained feature interventions and faithfulness tests	Error-preserving feature edits, complete-latent and raw-MLP controls

3 Methods

3.1 Model, prompts and activation capture

All experiments used `Qwen3-4B-Instruct-2507` in non-thinking mode with deterministic next-token evaluation. The model was loaded through Hugging Face Transformers with bfloat16 weights. Prompts were tokenised directly rather than wrapped in a chat template. Python, NumPy and PyTorch seeds were fixed to 787 for dataset order, train-validation splitting and initial SAE training. The selected transformer layers were {4, 8, 12, 16, 20, 24, 28}.

Three deterministic synthetic corpora were generated. The arithmetic corpus contains 1,000 two-digit sums, balanced over the 100 pairs of ones digits. The unit corpus contains five quantities, 20 contextual objects and ten templates, giving 1,000 prompts. The capitals corpus expands 189 country and state examples to 1,000 prompts by substituting non-capital cities from a fixed list. A forward hook recorded the MLP output at the final input token, producing one 1000×2560 matrix per selected layer and behaviour. Separate SAE sets were trained for arithmetic, units and capitals.

Arithmetic was evaluated autoregressively at the tens digit. For example,

Question: What is 58 + 83? **Answer:** 1

teacher-forces the hundreds digit of 141. The correct next token is 4; 3 is the counterfactual produced by dropping the carry from $8 + 3$. Matched no-carry sources preserved token length and the teacher-forced hundreds digit. An additional $44+83=127$ source distinguished an abstract no-carry state from source-specific transfer of the next digit.

Unit interventions operate on first-token answer prefixes because the complete answers are multi-token. The principal output contrast is

$$G_{\text{unit}} = \ell(\text{new}) - \ell(\text{j}), \quad (7)$$

where the prefixes represent newtons and joules. Force and mass prompts act as activation donors to an energy target. The mass donor is a matched factual recall control: it changes the physical quantity but should not transfer the newtons preference. Capital interventions analogously compare first-token prefixes for source and target cities.

3.2 SAE training and diagnostic selection

For each layer, a shared seeded 80:20 split supplied 800 training and 200 validation vectors. Activations were divided by

$$s_\ell = \frac{\mathbb{E}_{\text{train}} \|x^\ell\|_2}{\sqrt{2560}}, \quad (8)$$

so the mean norm of the normalised training vectors was approximately $\sqrt{2560}$. Each SAE had input width 2,560 and latent width 8,192. The initial ReLU models were trained for 75 epochs with Adam, learning rate 10^{-3} , batch size 64 and

$$\mathcal{L}_{\text{ReLU}} = \text{mean} [(u - \hat{u})^2] + 3 \times 10^{-4} \text{mean}(|z|). \quad (9)$$

The checkpoint with minimum validation loss was retained. One SAE contains 41,953,792 trainable parameters; “lightweight” therefore describes the runtime and data scope, not a low parameter count.

After the initial interventions, all three ReLU sets were evaluated on their fixed validation splits using FVE, mean non-zero latent count (L_0), cosine similarity and dead-feature fraction. The diagnostic motivated a pre-specified TopK comparison for arithmetic and units. Candidate values $k \in \{128, 256, 512\}$ were fixed before any candidate intervention result was examined. These models used the same layer set, width, data split and 75-epoch schedule, but Equation 4, unit-normalised decoder columns and no ℓ_1 term. A candidate was eligible only if mean validation FVE was at least 0.90, every layer’s FVE was at least 0.85, and mean dead-feature fraction was at most 0.80. Among eligible candidates, the model with the smallest mean validation L_0 was selected. Intervention outcomes were not inputs to this selection rule.

3.3 Contrast attribution graphs

At graph construction time each SAE was inserted at the final prompt position with a clean-value straight-through splice,

$$x' = x + \hat{x} - \text{stopgrad}(\hat{x}). \quad (10)$$

The forward value is exactly x , while gradients from the output pass through the SAE. This prevents reconstruction error from changing the baseline before features are attributed.

For target token t and contrast token c , the graph objective is

$$J = \ell(t) - \ell(c). \quad (11)$$

The first-order attribution of active feature i is $A_i = z_i \partial J / \partial z_i$. Between adjacent selected layers, the edge quantity is

$$E_{i \rightarrow j} = z_i \frac{\partial z_j}{\partial z_i} \frac{\partial J}{\partial z_j}. \quad (12)$$

Input-token edges use the analogous embedding dot-gradient score. The graph retains the 20 active nodes of largest absolute attribution per selected layer and edges above magnitude 10^{-6} . Decoder directions are projected through the unembedding matrix to obtain three diagnostic output labels. These labels provide evidence about a direction’s output alignment; they are not treated as proof that a latent is monosemantic.

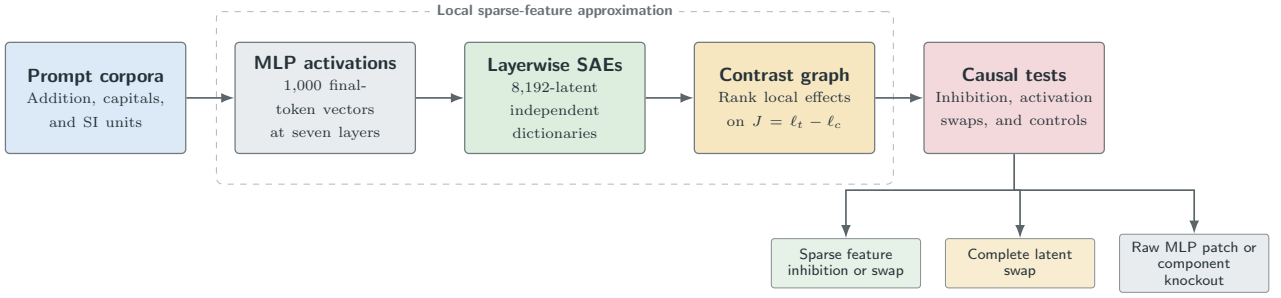
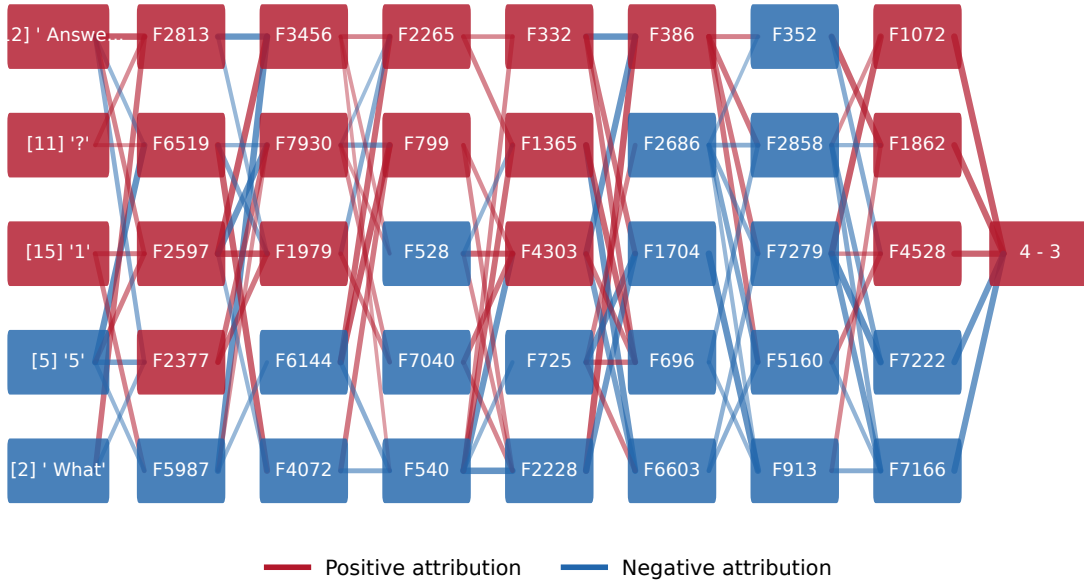


Figure 1: Independent small-scale reproduction pipeline. The lower branch distinguishes sparse feature tests from broader positive controls.

Backward-connected visual subset of the carry attribution graph Logit diff: '4' - '3'

Input token Layer 4 Layer 8 Layer 12 Layer 16 Layer 20 Layer 24 Layer 28 Logit contrast



Displayed subset: up to 5 nodes per stage and 14 edges per transition; complete graph retained in JSON/HTML.

Figure 2: Backward-connected visual subset of the arithmetic attribution graph for $\ell(4) - \ell(3)$. Up to five nodes per stage are shown; the complete pruned graph is retained in JSON and interactive HTML form. Red and blue denote positive and negative local attribution respectively.

3.4 Causal interventions and confirmation protocols

Feature interventions preserve the SAE reconstruction error. If $z^{(-S)}$ sets coordinates S to zero, inhibition edits the true MLP output by only the decoded feature difference,

$$x' = x + s_\ell W_d \left(z^{(-S)} - z \right). \quad (13)$$

A source-to-target swap applies

$$x'_t = x_t + \alpha s_\ell W_d (z_s - z_t), \quad (14)$$

either on a selected coordinate set or on the complete 8,192-dimensional code. Raw MLP patching replaces selected target MLP outputs directly and bypasses the SAE. Component knockout sets an MLP, attention or block output to zero. The latter two are positive or localisation controls, not sparse-circuit evidence.

Early diagnostics compared final-position and all-position edits. Token IDs and pre-edit MLP norms were printed to verify the actual positions and non-zero activation being modified. Because SAEs were trained at the final position, the final feature screens use only that position; all-position interventions are reported explicitly as distribution-mismatched exploratory controls.

Two initial discovery–confirmation protocols were frozen. For arithmetic, features from one carry graph were ranked on eight discovery pairs by the target intervention effect minus the effect on a matched no-carry control. The predeclared top-ten panel was then tested on 24 disjoint pairs. Success required a negative mean carry-minus-control difference, a bootstrap 95% interval wholly below zero, and a negative mean carry-target effect. For units, 73 positive-attribution features from one force graph were ranked on eight physical systems by the force-source swap effect minus the matched mass-source effect. The frozen top-ten panel was tested on 16 disjoint systems. Success required both the force effect and force-minus-mass effect to be positive with bootstrap intervals wholly above zero. All intervals use 5,000 seeded non-parametric resamples of the evaluation cases and quantify variation across prompt systems, not model-sampling uncertainty.

A final arithmetic protocol addressed a remaining confound: a feature may separate the correct and dropped-carry tokens because it encodes the output digit rather than carry status. After excluding 44 previously inspected arithmetic pairs, 105 fresh pairs met the clean-baseline criteria. Thirty-two pairs were assigned to discovery and 32 to confirmation while balancing shared tens-digit strata. At each layer, a linear raw-MLP direction was fitted after subtracting output-digit-specific centres. All 57,344 TopK-256 latents were then ranked on discovery data by their standardised carry-minus-no-carry activation difference within output-digit strata. Confirmation data did not affect ranking. The top-ten panel was the predeclared primary; cumulative panels of 1, 3, 5 and 20 features were secondary. After the top-ten primary failed, the observed top-20 effect was treated as hypothesis generation rather than substituted as the primary result. Its exact feature IDs were frozen in code and evaluated once on 32 of the 41 remaining baseline-qualified pairs that had received no feature intervention. Replication cases were selected using only baseline eligibility and output-digit balance. The stopping rule required a negative carry-target mean and a carry-minus-control bootstrap interval wholly below zero; no further panel or seed was permitted after this test.

The first units baseline-only screen used indirect wording and left too few fully qualified systems. Before any feature intervention was run, the protocol was amended to direct prompts of the form “the official

Table 2: Initial baseline audit. “In top- k ” denotes an acceptable answer in the saved top-20 or prefix-aware completion check.

Behaviour	Prompts	Top-1 correct	In top- k
Addition	200	200 (100.0%)	200 (100.0%)
SI units	200	134 (67.0%)	193 (96.5%)
Capitals	189	94 (49.7%)	158 (83.6%)

SI unit used to measure [quantity] is named”. Four wording variants were audited for 64 physical systems; eligible direct prompts were then split by system. The exact discovery and confirmation prompts were absent from the SAE corpus, and no physical system appeared in both splits.

3.5 Execution and reproducibility

The implementation is maintained in the public repository https://github.com/evey-dev/test_run. Dataset generation, activation capture, SAE training, graph construction, intervention, diagnostics and figure generation are standalone Python modules under `src/`; the notebooks are orchestration layers rather than the sole executable form of the analysis. YAML files record model and SAE settings, and all principal results are serialised as JSON or CSV before plotting.

GPU execution used Google Colab after CSD3 became unavailable during the experimental period. Each final notebook mounts Google Drive, pulls the current repository from GitHub, installs the package in editable mode, restores or trains checkpoints, and copies checkpoints and outputs back to Drive. Recorded notebook metadata and logs identify NVIDIA T4 and A100-SXM4-80GB runtimes; no claim is made that results require either GPU type. The model runs in bfloat16, whereas aggregate statistics and bootstrap calculations use NumPy precision. The final paths are documented in `README.md`: arithmetic TopK training and screening use `run_gpu_math_topk_retrain.ipynb`, `run_gpu_math_carry_feature_followup.ipynb` and `run_gpu_math_carry_balanced_localization.ipynb`; units use `run_gpu_units_topk_retrain.ipynb`; and retained capital experiments use `run_gpu_capitals_final.ipynb`. Earlier notebooks remain as an auditable record of exploratory settings rather than being silently replaced.

4 Results

4.1 Baseline competence

Table 2 reports the retained initial audit. Arithmetic was reliable, whereas units and capitals included multi-token answers and templates that sometimes elicited an intervening word rather than the answer prefix. Every causal benchmark therefore reapplied a clean-baseline eligibility rule to each specific prompt. Aggregate baseline percentages are competence checks from an earlier prompt-set version, not estimates for the final confirmation pools.

4.2 Stage I: the original ReLU representation

The initial attribution graphs produced only small sparse effects. The most important diagnostic question was whether this reflected poor SAE reconstruction. Table 3 and Figure 3 reject that simple explanation. Mean validation FVE was 0.954 for arithmetic, 0.959 for units and 0.907 for capitals.

Table 3: Original ReLU SAE diagnostics averaged over seven layers on the fixed 200-vector validation splits. Dead fraction is the fraction of dictionary features inactive on the validation split.

Domain	Mean validation FVE	Mean validation L_0	Mean dead fraction
Addition	0.9545	2,166.6	0.0106
SI units	0.9591	3,544.6	0.0000
Capitals	0.9068	3,102.8	0.0000

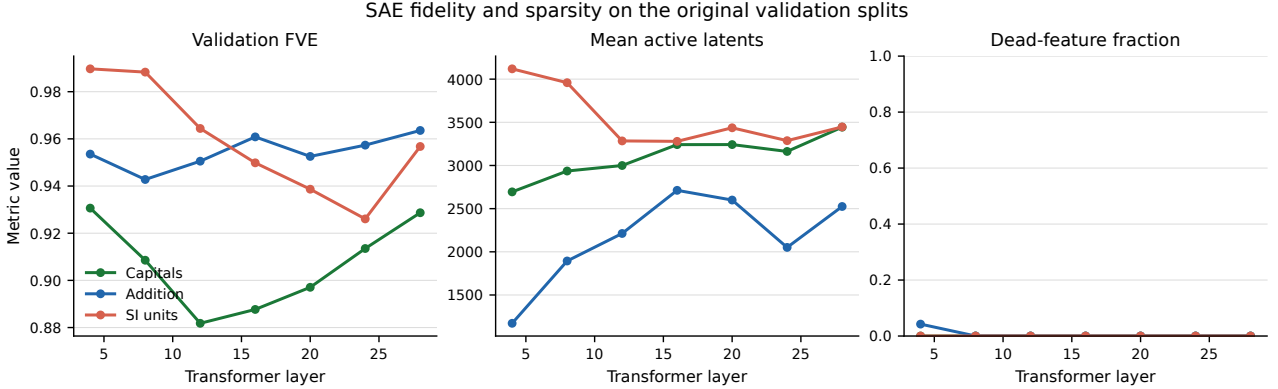


Figure 3: Layerwise fidelity and achieved sparsity of the original ReLU SAEs. High FVE coexists with thousands of active latents. The plotted dead-feature fraction is measured on combined train and validation support.

Mean cosine similarity was at least 0.993 in every domain. However, the corresponding codes activated 2,167, 3,545 and 3,103 latents per example: approximately 26%, 43% and 38% of the dictionary. A representation with thousands of active coordinates is reconstructive but does not provide the compact bottleneck needed for an Anthropic-like feature intervention.

Broad interventions were introduced to test whether the model locations still contained causal information. For the unit minimal pair, an all-position complete-latent force-to-energy patch changed the top prefix from *j* to *new*: $p(\text{new})$ rose from 0.0240 to 0.4980 while $p(j)$ fell from 0.7969 to 0.2354. Swapping the 76 positive graph features increased the *new* logit by only 0.37 and did not flip the answer. On six baseline-qualified graph-held-out pairs, final-position sparse swapping increased G_{unit} by a mean 0.229 logits (95% interval [0.146, 0.333]), with the predicted direction in all six cases but no top-token transfer. These findings localised a difference between information in the overall SAE code and control by its graph-selected coordinates.

Arithmetic showed the same hierarchy. On 58+83, inhibiting all 73 positive carry-graph features reduced the correct-minus-dropped-carry gap from 7.12 to 6.63. A strength-two complete-latent patch from 54+83 reduced it to 0.74, and a raw all-position MLP patch made 3 the top token. The apparently successful transfer was not yet evidence of an abstract carry state: an alternative 44+83 source made its own next digit, 2, the top token under the same broad patch. On 12 graph-held-out carry pairs, raw MLP patching changed the gap by -13.99 logits and transferred the source top token in all cases, whereas ReLU sparse inhibition changed it by only -0.396 ($[-0.458, -0.333]$). The raw control establishes causal sufficiency of the selected MLP state, but its source-digit dependence limits the semantic claim.

An all-position MLP knockout supplied a complementary diagnostic. Across the seven selected layers,

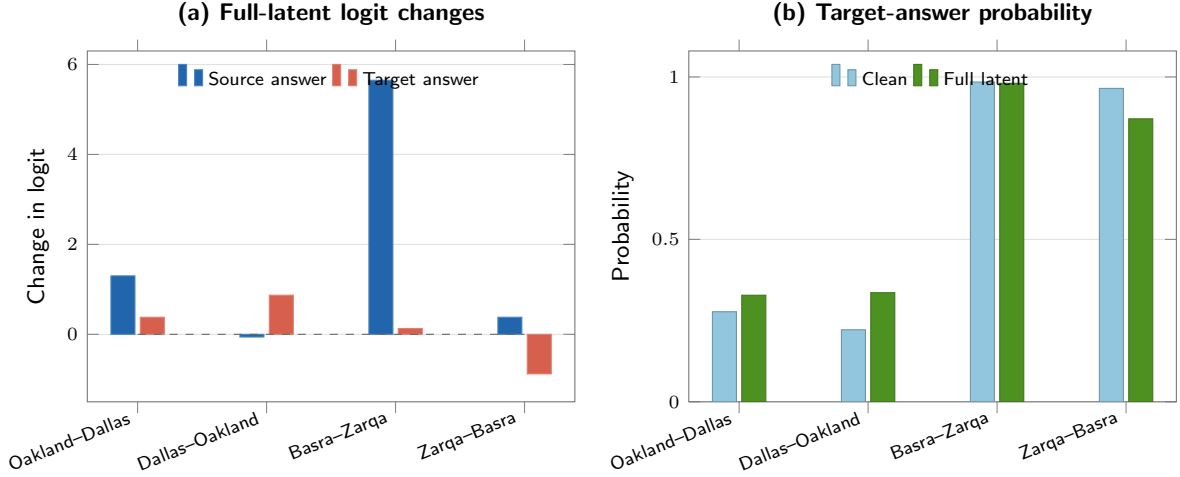


Figure 4: Complete-latent capital-city interventions. Increasing clean confidence from the Dallas/Oakland pair to Zarqa/Basra did not yield answer transfer.

4 remained top but its probability fell from approximately 1.0 to 0.6406; 3 rose to 0.1836 and 2 to 0.1426. Layer 24 alone was particularly destructive and made 2 the top prediction. This demonstrates that layer 24 contributes strongly to answer selection. It does not identify a carry representation, because multiple alternative digits increased and zeroing an entire component is not a concept-specific intervention. Attention knockout was weaker, while full-block knockout produced non-answer tokens and was consequently excluded as an interpretable circuit test.

4.3 Stage II: capital-city controls

Capital-city experiments tested whether weak transfer was merely a consequence of uncertain clean predictions. In the initial Dallas/Oakland pair, **Austin** had probability 0.2773 for Dallas, while **Sacramento** had probability 0.2217 for Oakland and was not the top token. Complete-latent patches moved the relevant logits but did not exchange the answers. A second pair therefore used high-confidence baselines: Zarqa predicted the Amman prefix with probability 0.9844 and Basra predicted Baghdad with probability 0.9648. Patching Basra into Zarqa raised the Baghdad logit from 6.97 to 12.62, but its probability remained 0.0002 and Amman remained top. Graph-selected swaps were smaller.

The high-confidence control removes one explanation for the negative result, but not all of them. Country/state knowledge may be represented across components not captured by final-position MLP SAEs, and first-token patching may not reproduce the multi-token retrieval trajectory. The result is therefore reported as a negative factual-recall reproduction rather than evidence that the model lacks structured geographic knowledge.

4.4 Stage III: TopK arithmetic and failed initial specificity tests

The ReLU diagnostic motivated a representation-level correction rather than repeatedly enlarging a dense feature set. Arithmetic TopK candidates preserved high fidelity while reducing activity by approximately an order of magnitude. TopK-128 achieved mean FVE 0.9466 and mean $L_0 = 124.6$, but its mean dead-feature fraction of 0.832 exceeded the predeclared threshold. TopK-256 was therefore selected with mean FVE 0.9410, minimum layer FVE 0.9319, mean $L_0 = 171.6$ and dead fraction 0.716.

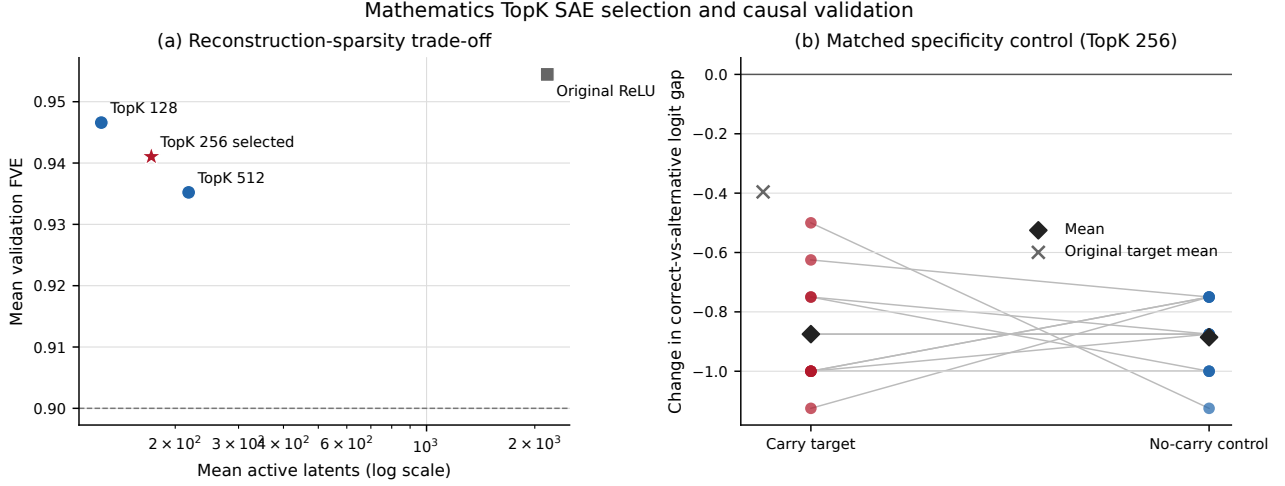


Figure 5: Predeclared arithmetic TopK selection and matched causal test. The selected representation is substantially sparser, but its graph-feature effect is equally present on no-carry controls. Lines in panel (b) connect matched prompt pairs.

TopK-512 was eligible but denser and was not selected.

On 12 graph-held-out all-position pairs, inhibiting the selected graph features reduced the carry-target gap by a mean -0.875 logits ($[-0.969, -0.771]$), more than twice the original ReLU mean. Every target moved in the expected direction. The matched no-carry gap, however, changed by -0.885 logits. The paired target-minus-control estimate was 0.010 ($[-0.125, 0.167]$), with the carry target more negative in only four of 12 pairs. The intervention therefore altered arithmetic digit confidence but was not selective for carrying.

Position mismatch could have concealed a final-token feature effect. Repeating the benchmark at the SAE training position produced a mean target change of -0.656 and control change of -0.625 ; the paired estimate remained indistinguishable from zero at -0.031 ($[-0.156, 0.104]$). Finally, features were ranked on eight discovery pairs and a frozen top-ten panel was evaluated on 24 disjoint confirmation pairs. The target gap decreased by -0.141 logits, but the no-carry control also decreased by -0.109 . The paired specificity effect was -0.031 ($[-0.089, 0.026]$), failing the predeclared rule. Twenty-one of 24 carry targets moved in the predicted direction, yet none changed top prediction. Five matched random panels produced paired means between -0.036 and 0.021 , reinforcing the conclusion that the residual top-ten effect was small relative to panel variability.

This sequence distinguishes sensitivity from mechanism. The TopK dictionary made interventions larger and the selected graph features consistently affected digit logits, but the matched control showed that the effect was generic to the arithmetic contexts tested. It would therefore be incorrect to label this graph-derived panel “carry 1” solely because it was found on a carry graph.

4.5 Stage IV: output-digit-balanced carry localisation

The failed graph screens left open a more precise hypothesis: carry information could be present in the SAE basis but obscured by answer-digit variation or absent from the single attribution graph. The balanced protocol therefore searched all 57,344 latents using discovery activations rather than graph membership. Output-digit-conditioned raw-MLP probes confirmed that the selected site contained carry information. Confirmation AUC was 1.000 at both layers 24 and 28, compared with 0.754 and

Final-position TopK SAE carry-feature discovery and confirmation

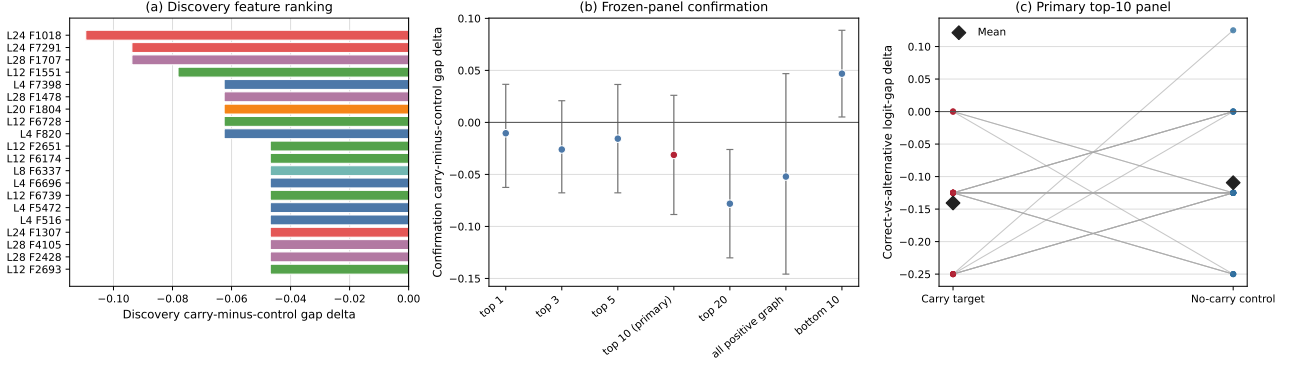


Figure 6: Final-position arithmetic discovery and frozen confirmation. Negative values favour carry selectivity. The primary top-ten interval crosses zero, so the panel does not satisfy the predeclared carry-specificity criterion.

0.800 at layers 4 and 8 and values below 0.5 at layers 12–20. The activation-ranked top-20 SAE panel comprised 11 layer-24 and nine layer-28 latents. On the first 32-pair confirmation set, its mean within-digit standardised activation difference was 0.967 ([0.866, 1.077]).

The predeclared top-ten primary was retained as a failed result. Although its confirmation activation AUC was 1.000 and its mean conditioned difference was 1.130 ([1.011, 1.253]), inhibition changed the carry-target gap by only -0.008 logits and the no-carry gap by -0.023 . The paired specificity estimate was therefore $+0.016$ ($[-0.035, 0.063]$). This directly demonstrates that perfect held-out decodability need not identify a causally selective feature set.

The secondary top-20 panel produced a different pattern on the same confirmation set: carry-target inhibition changed the gap by -0.078 logits, the no-carry control changed it by $+0.023$, and the paired effect was -0.102 ($[-0.148, -0.051]$). Because top-20 was secondary, this interval was treated only as hypothesis-generating. The exact 20 IDs were then frozen without reranking and tested once on 32 intervention-untouched eligible pairs. In this case-level replication, the carry-target mean was -0.059 ($[-0.090, -0.023]$), the no-carry mean was $+0.027$ ($[-0.004, 0.059]$), and the paired specificity effect was -0.086 ($[-0.137, -0.039]$). The criterion passed; 62.5% of pairs had a more negative carry effect than control, and neither condition changed the top prediction.

Figure 7 supports a bounded positive claim. Inhibiting a distributed late-layer SAE panel selectively reduced the preference for the carry-consistent tens digit relative to a matched no-carry control, and the effect recurred on cases not used for feature ranking or the first intervention test. The magnitude remained small, no output token transferred, and the experiment reused one SAE basis. It therefore identifies a carry-associated causal panel at the tested MLP sites, not a single carry bit, a complete arithmetic circuit, or a basis-stable mechanism across training seeds.

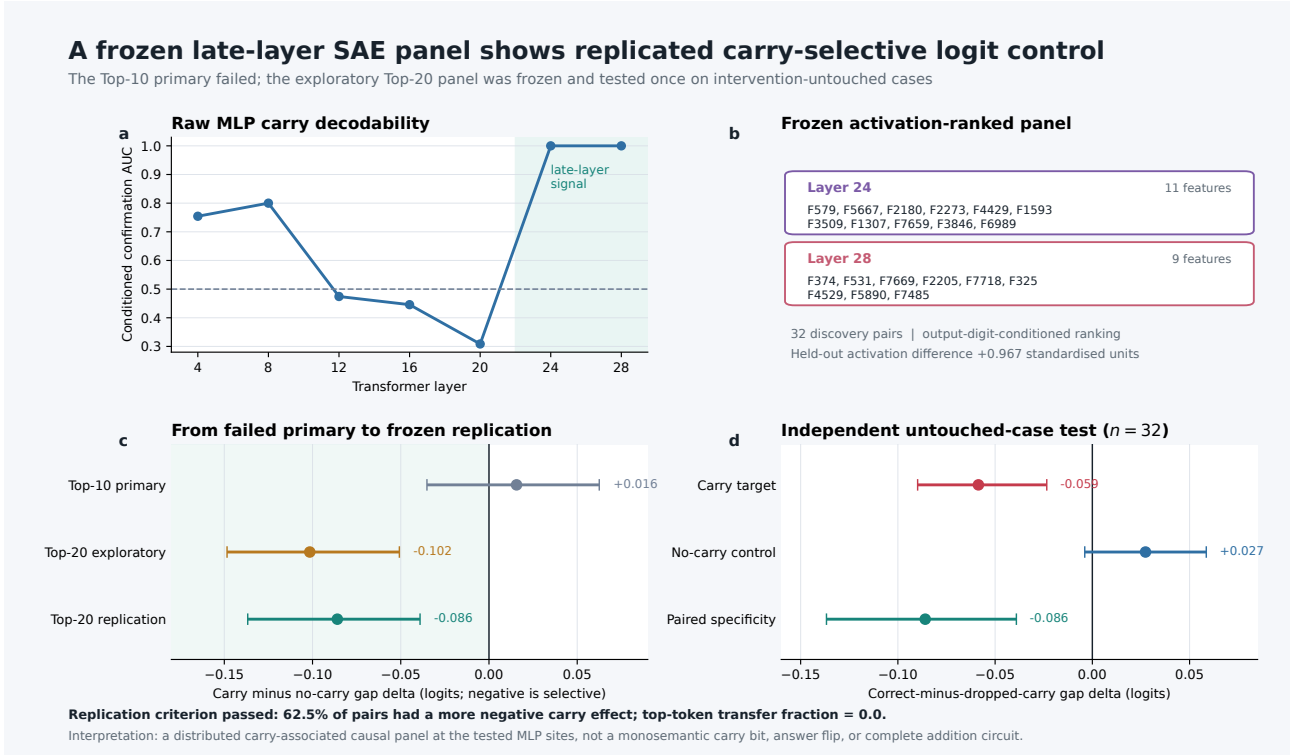


Figure 7: Output-digit-balanced arithmetic localisation and case-level replication. Panel (a) reports confirmation AUC for carry directions fitted in raw MLP outputs after output-digit centring. Panel (b) lists the exact activation-ranked top-20 panel frozen after discovery. Panel (c) preserves the failed top-ten primary, distinguishes the secondary top-20 observation from its untouched-case replication, and reports paired carry-minus-control effects. Panel (d) decomposes the replication into target and matched-control changes. Error bars are seeded bootstrap 95% intervals across prompt pairs.

4.6 Stage V: a compact units panel

The same fixed TopK procedure was applied to units without changing the original SAE corpus. All three candidates met the diagnostic thresholds. TopK-128 was selected because it had the smallest mean validation activity: FVE 0.9358, minimum layer FVE 0.8891, mean $L_0 = 99.8$ and dead fraction 0.799. TopK-256 and TopK-512 had mean L_0 values 132.7 and 165.5 respectively. Relative to the original units ReLU SAE, the selected code used 35.5 times fewer active latents while losing only 0.0233 mean FVE.

An initial graph-held-out benchmark provided an independent directional check. Eight of 16 candidate prompt pairs met both clean source and target criteria. Across those eight, swapping all positive graph features at the final token increased G_{unit} by 0.586 logits ($[0.352, 0.883]$), with the predicted direction in every case. The complete latent and raw MLP controls produced changes of 1.352 and 1.617 logits. No condition changed the top prediction. Compared with the original ReLU sparse effect of 0.229 on its six eligible pairs, TopK provided a larger but still broad graph-set effect.

The frozen feature screen then tested whether that effect could be concentrated and distinguished from another unit donor. Of the 73 positive graph features, the ten highest-ranked discovery features comprised eight layer-28 features, one layer-8 feature and one layer-4 feature. Six decoder directions had force-related top-output labels; this is supporting evidence, not a semantic definition. On 16 disjoint confirmation systems, the force-source swap increased the newtons-minus-joules gap by 1.234 logits ($[1.148, 1.332]$). The mass-source control changed it by -0.090 ($[-0.129, -0.051]$), giving a paired specificity effect of 1.324 ($[1.223, 1.430]$). All 16 systems had a positive force effect and a force effect larger than the corresponding mass effect.

The cumulative panel response was coherent: force-minus-mass effects were 0.352, 0.813, 1.055 and 1.324 logits for the top 1, 3, 5 and 10 features. The top-20 panel gave 1.348 and all 73 positive graph features gave 1.391, whereas the reverse-ranked bottom-ten control gave only 0.055 ($[0.008, 0.098]$). Five layer-count-matched random panels had effects between 0.328 and 1.191; the primary panel exceeded all five, although five draws are insufficient for a formal randomisation test. The complete-latent and raw-MLP paired effects were 2.840 and 2.891 logits respectively.

The clean mean confirmation gap was -9.41 , the force-swapped gap was -8.17 and the mass-control gap was -9.50 . Joules consequently remained the top prediction in every case. The force-source effect of the top-ten panel was 97.5% of the corresponding 73-feature force-source effect and 40.6% of the complete-latent force-source effect. These are ratios of observed intervention effects, not fractions of variance or of the mechanism explained.

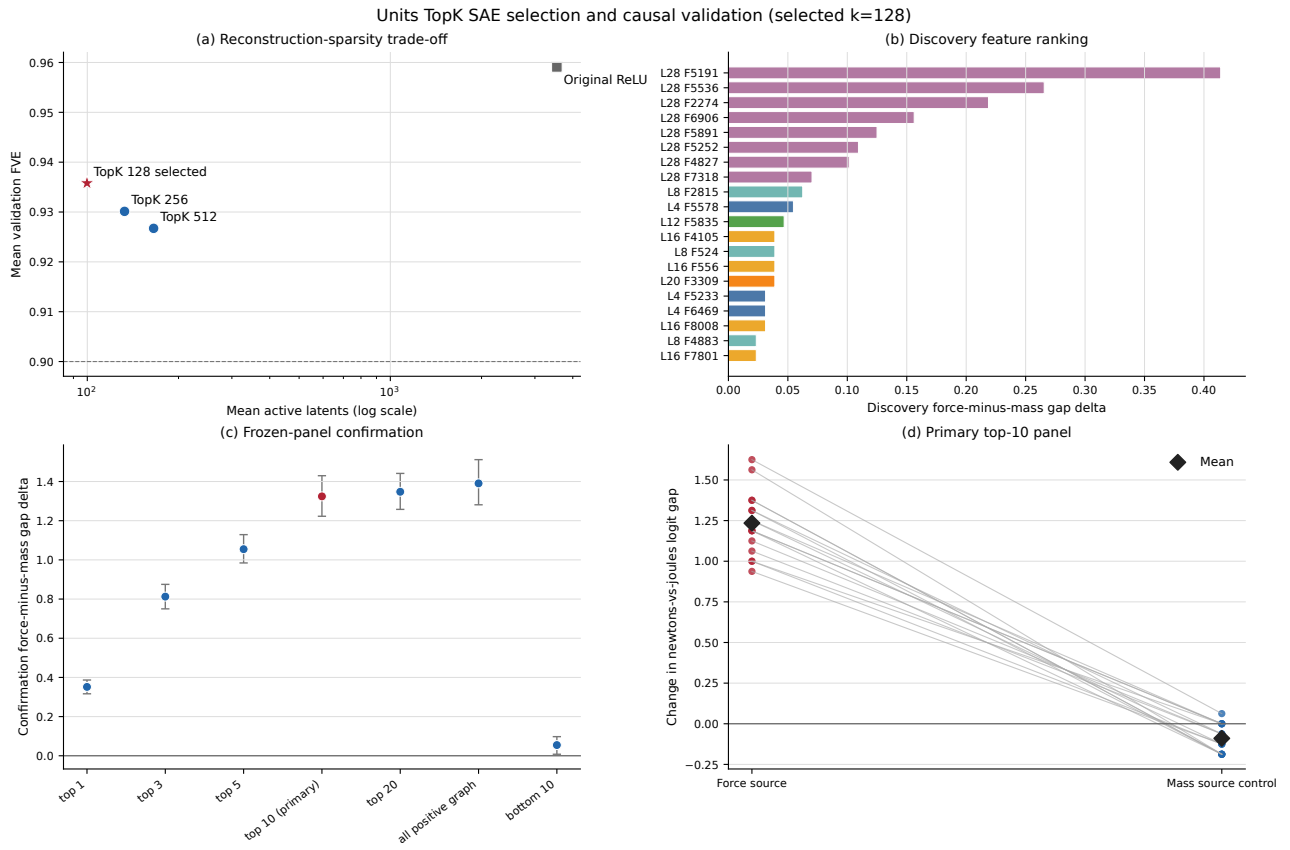
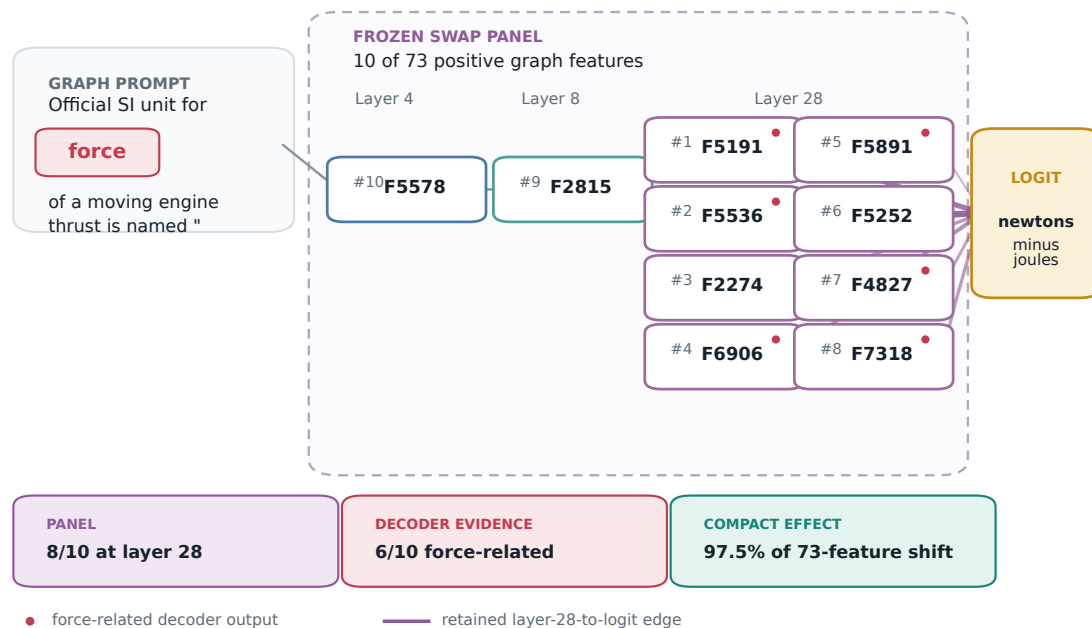


Figure 8: Units TopK selection, discovery ranking and frozen confirmation. The candidate was selected from reconstruction diagnostics before interventions; the top-ten panel was ranked only on discovery systems. In panel (d), every confirmation system shows a larger shift for force than for the mass control.

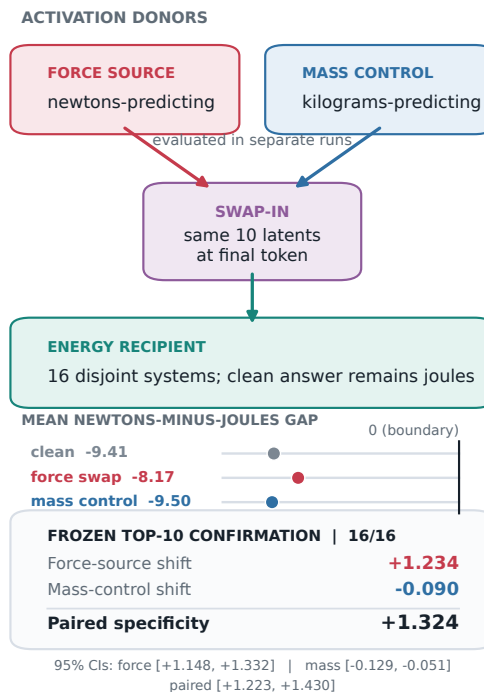
Compact force-to-unit transfer in Qwen3-4B-Instruct

Attribution-guided TopK SAE panel and frozen swap-in confirmation

A Attribution-guided feature panel



B Frozen causal confirmation



Interpretation: the frozen panel produced a selective logit shift, not a token flip. Its force-source effect was 40.6% of the full-latent swap effect. Effect ratios are intervention comparisons, not variance explained.

Solid layer-28 arrows are retained graph edges. The dotted path compresses unshown graph nodes; the diagram is an evidence summary rather than a complete computational circuit.

Figure 9: Evidence summary for the compact force-associated panel. Feature IDs, layer assignments and displayed layer-28 edges are taken from the retained graph; swaps replace the ten final-token latent values with source values. The diagram summarises a selective logit shift, not a token flip or a complete computational circuit.

This result satisfies the predeclared causal criterion and is the strongest partial reproduction in the study. Its defensible interpretation is that a small, attribution-guided, layer-28-dominated group carries force-associated information that selectively shifts unit-token preference across new physical systems. The experiment does not show that these ten latents are uniquely necessary, that each is monosemantic, or that the panel alone computes factual recall.

5 Discussion

5.1 What was and was not reproduced

The project reproduced the complete experimental workflow requested for a small-scale study: fixed behavioural baselines, MLP activation capture, behaviour-specific per-layer SAEs, a pruned input–feature–logit graph, and causal validation in the underlying model. It recovered two bounded sparse effects with different inferential histories. The units top-ten panel was selected under a predeclared protocol and generalised from one graph prompt through a separate discovery set to 16 disjoint confirmation systems, where it was specific relative to a mass donor. In arithmetic, the predeclared top-ten primary failed, but an activation-ranked top-20 secondary hypothesis subsequently reproduced under a frozen one-shot test on 32 intervention-untouched cases.

Both results remain materially weaker than the target paper’s circuit narratives. They change logit contrasts without changing answers; only one SAE seed was tested; the arithmetic panel was selected after a failed smaller primary; and five units random panels do not define a precise null distribution. The terms “carry-associated panel” and “force-associated panel” are therefore more accurate than “carry circuit” or “force circuit”. The evidence diagrams deliberately expose these boundaries.

Arithmetic supplies both controlled failures and a limited positive result. Graph-derived features influenced digit logits after TopK retraining but did not distinguish carry targets from no-carry controls, while broad patches transferred source-specific answer state. Output-digit conditioning then isolated a later-layer signal whose 20-feature causal effect replicated across new cases. The contrast shows why concept-matched controls and held-out replication are necessary before assigning semantic meaning to a feature group. Capitals remained a negative reproduction after clean confidence was increased, leaving factual-recall transfer less consistent across domains.

5.2 Why reconstruction was insufficient

The diagnostic evidence permits a mathematical explanation without attributing all negative results to generic “SAE limitations”. First, the original ℓ_1 objective is scale-degenerate when decoder norms are unconstrained. For $0 < c < 1$, the transformation

$$W'_e = cW_e, \quad b'_e = cb_e, \quad W'_d = c^{-1}W_d \quad (15)$$

gives $z' = cz$ by positive homogeneity of ReLU and leaves $W'_d z' = W_d z$ unchanged, while multiplying the latent ℓ_1 penalty by c . This does not prove that optimisation fully exploited the degeneracy, but it proves that the raw penalty is not an invariant measure of sparsity. The observed L_0 values confirm the practical consequence: high FVE was achieved with dense codes. TopK plus decoder normalisation removes this particular ambiguity by controlling support size directly, and the units result improved accordingly.

Second, reconstruction, decodability and causal separability are different objectives. Equation 3 constrains average clean-distribution error $\|x - \hat{x}\|$, whereas a feature intervention asks whether a selected Δz produces the intended and concept-specific change $\Delta x = W_d \Delta z$. For output objective J , a local expansion is

$$J(x + \Delta x) - J(x) = \nabla J(x)^\top \Delta x + O(\|\Delta x\|_2^2). \quad (16)$$

Low reconstruction error does not bound the second-order term for edited points, nor require that semantically distinct factors occupy distinct coordinates. Correlated latents can reconstruct well while finite ablation or swap effects are distributed across many directions. The balanced arithmetic result makes the distinction direct: the top-ten activation score classified held-out carry status perfectly after output-digit conditioning, yet its causal specificity interval crossed zero. Selective control appeared only after expanding to 20 frozen latents and then survived a separate case-level test.

Third, independent per-layer output SAEs do not model cross-layer feature effects in the manner of Anthropic’s CLT. A CLT can assign one upstream feature decoder contributions to several later MLPs, whereas the present graph links separately trained dictionaries through local derivatives. The clean-value splice preserves the unperturbed output, but it does not create explicit error nodes or guarantee perturbation faithfulness. The target methods paper likewise treats faithfulness as an empirical property rather than a consequence of graph construction [2].

5.3 Limitations of the experimental scope

The lightweight design imposed statistical and representational constraints. Each approximately 42-million-parameter SAE was fitted to 800 training vectors from a highly templated domain. Validation vectors share the same generator, so high FVE does not establish dictionary uniqueness or generalisation to arbitrary language. A single seed provides no estimate of feature-basis stability. Although the final units confirmation prompts were absent from the SAE corpus, they remain synthetically related to it.

The SAEs were trained only on final-token MLP outputs. Early all-position edits therefore encoded token positions outside their training support, and the arithmetic teacher-forced digit was not the same endpoint used in the original activation corpus. The final screens corrected the intervention-position mismatch but did not retrain on teacher-forced arithmetic states. Attention and residual-stream features were not decomposed. A weak MLP-feature result therefore cannot establish that a mechanism is absent elsewhere in the model.

The confirmation sample sizes remain modest: 24 pairs in the graph-derived arithmetic screen, 32 pairs in each balanced arithmetic intervention set, and 16 unit systems. Bootstrap intervals condition on these constructed prompt pools and do not account for SAE seed, graph seed or prompt-generator uncertainty. The arithmetic replication is independent at the case level but reuses the same SAE weights and feature IDs, so it does not test dictionary stability. It also does not compare the 20-feature panel with size-matched random panels; it tests persistence of the exact selected panel, not its extremeness among arbitrary panels. Unit and capital measurements use first-token prefixes rather than probabilities of complete answer strings. Finally, bfloat16 logits are quantised visibly in increments such as 0.0625 or 0.125; effects near that scale require aggregation and replication across cases, rather than interpretation from an isolated prompt.

One implementation defect was identified during audit: an exploratory raw-MLP layer-scan path patched every requested layer in each nominal single-layer row. Those rows are excluded from all

layer-localisation claims, and the hook logic has been corrected. Aggregate raw patches, component knockouts, SAE layer scans and the final discovery–confirmation experiments use different paths and are unaffected. Reporting and excluding this defect is part of the reproducibility analysis rather than a post hoc adjustment to the main effect.

5.4 Computational and infrastructure constraints

The intended CSD3 environment became unavailable to the project for approximately two weeks during a late-June heatwave. A University notice attributed the major incident to cooling failure during recent hot weather and listed CSD3 among the unavailable Research Computing Data Hall services [11]. The CSD3 status page subsequently recorded the service as unavailable during post-incident recovery [10]. The project’s unbacked activation matrices and checkpoint artifacts were no longer available and had to be regenerated. This is consistent with CSD3’s documented storage model: active `hpc-work`/RDS storage is not backed up, whereas node-local storage is deleted after jobs [12]. Source code remained versioned in GitHub, which made the pipeline recoverable but did not eliminate the cost of recapturing activations and retraining SAEs.

The replacement Colab environment permitted completion of the targeted experiments but changed the feasible search strategy. Colab meters paid GPU access against a compute-unit balance, and Google states that hardware availability, runtime limits and usage conditions vary over time and are not published as fixed limits [9]. Resource use is therefore described qualitatively rather than assigned a universal compute-unit-per-hour rate. Compute was prioritised for fixed candidate sweeps, held-out controls and frozen confirmation rather than all-layer, all-token activation corpora, multiple SAE seeds or a cross-layer transcoder. This constraint limits uncertainty estimates and architectural fidelity, but it does not explain away the observed contrast between arithmetic and units: both final TopK protocols were executed within the same repository and intervention framework.

5.5 Prioritised further work

The most informative next experiment is replication of the units top-ten effect across SAE and graph seeds with an enlarged, pre-generated confirmation pool. The complete selection procedure, including feature ranking and panel size, must be rerun independently; retaining the same feature IDs after retraining would not test basis stability. More layer-count-matched random panels would permit a formal randomisation comparison.

For arithmetic, the next priority is not another panel size in the current SAE. The full output-digit-conditioned discovery and exact top-20 replication should be repeated across independently trained SAE seeds, with feature ranking rerun from the beginning and a newly generated confirmation pool. SAEs should also be trained on the teacher-forced decimal positions that will be intervened upon, with carry status, source next digit and target next digit varied factorially. This would test whether the replicated case-level effect corresponds to a stable representational factor rather than one convenient basis.

At greater computational scale, per-layer transcoders or a small CLT should be compared with the independent output SAEs at matched reconstruction and active feature budgets. Explicit error nodes and perturbation-faithfulness tests would bring the graph semantics closer to the target work. Attention or residual stream decompositions may also be necessary for factual retrieval. These extensions are

motivated by specific diagnostic failures in the present study, not solely by the expectation that a larger model or dictionary must perform better.

6 Conclusion

This study independently reproduced a small-scale circuit-tracing workflow for an open language model and evaluated each link in the inferential chain. The original ReLU SAEs achieved high held-out reconstruction fidelity but activated thousands of latents. Broad patches established that the selected MLP states contained causal answer information, while matched controls exposed source-digit transfer in arithmetic and weak capital-city transfer. A predeclared TopK correction substantially improved sparsity and intervention magnitude. The first arithmetic graph and top-ten tests remained non-specific; an output-digit-balanced extension subsequently identified a frozen 20-feature panel whose small carry-selective inhibition effect reproduced on 32 untouched cases. The units analysis independently yielded a stronger frozen ten-feature panel whose force-source effect generalised to 16 disjoint systems and was absent under a mass-source control.

The resulting conclusion is a partial but substantive reproduction. Sparse SAE features can provide compact causal control in this lightweight Qwen setting, but reconstruction quality, decodability and directional logit movement are insufficient on their own. Mechanistic interpretation requires concept-matched controls, held-out confirmation, honest treatment of adaptive hypotheses, and explicit separation of sparse evidence from broad activation patching. The units panel satisfies a predeclared criterion for a force-associated shift; the arithmetic panel satisfies a later frozen case-level replication criterion for a smaller distributed carry-associated shift; and capitals define where transfer remained unsuccessful. This combination of positive and negative evidence is the principal reproducibility result.

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Declaration of use of autogeneration tools

Generative AI tools, including OpenAI Codex, were used during software development to assist with code review, debugging, experimental documentation, figure generation, and the structuring and drafting of this report. GPU experiments were executed by the candidate. The candidate remains responsible for checking the final analysis, numerical claims, citations, wording and conclusions. This declaration describes the use of these tools in the preparation of the submitted work.